THIS TORRY HARRIS

Case study

Gen AI augmented customer care agent

Real-time GenAI augmentation to improve customer care outcomes

www.torryharris.com/ae-en

Who

A leading provider of business process services.

What

A Gen AI augmented customer care agent that supports live customer care interactions with real-time contextual assistance, knowledge retrieval, and next-best guidance.

How

Torry Harris Integration Solutions provided a modular, productized solution that combines:

- Real-time conversation transcription
- Context retrieval from enterprise knowledge and backend systems
- LLM-driven response generation and recommendations
- An admin console and dashboards for governance and improvement

Results

Building on the data generated in the runtime environment, the solution delivered:

- Increased agent efficiency by **40%**
- AHT (Average Handle Time) reduced by **50%**
- Agent Utilization increased by **25%**
- Resolution Rate increased by **45%**
- Transfer Rate reduced by **70%**

Copyright © 2026 Torry Harris Business Solutions FZ - LLC

The business problem

Customer care agents often need to navigate multiple sources of information while handling live calls. This increases handle time, creates inconsistencies in customer responses, and increases training burden and operational costs. Transfers between teams further slow resolution and reduce customer satisfaction.

The solution

Torry Harris Integration Solutions implemented a GenAI-powered Augmented Customer Care Agent to support agents during live calls. The assistant listens to conversation context, retrieves the most relevant knowledge and customer context, and provides real-time answers and recommended actions, reducing the cognitive load on agents and improving consistency.

What is the Gen AI augmented customer care agent?

A real-time AI assistant that augments human agents during live calls by:

- Monitoring conversation flow
- Surfacing relevant knowledge instantly
- Generating grounded responses and recommended actions
- Helping resolve issues faster and more consistently

Copyright © 2026 Torry Harris Business Solutions FZ - LLC

How the customer care experience was transformed

The call handling workflow was redesigned so that the human agent remains the primary interface, while the Augmented Agent continuously provides support in real time.

Approach: Augmented agent – Solution framework

1. Customer contacts support
2. Call is received by an agent
3. Voice stream converted to text via speech-to-text
4. AI engine retrieves knowledge + backend context via APIs
5. LLM generates suggested responses and next-best guidance
6. Agent resolves the issue faster with AI assistance

Telephony Gateway

Assisting the Agent

- Input Voice stream acquisition from call interface
- Realtime Speech to text engine
- Text-based AI knowledge base conversational interface (API)
- API interface to access customer context and product information
- Outcome Text stream sent to the web interface

Automatically transcribes the call for the agent

Core components

- Telephony/call interface
- Real-time speech-to-text engine powered by **Microsoft Azure Speech to Text**
- AI orchestration layer (retrieval + grounding + prompting)
- LLM response engine powered by **Microsoft Azure OpenAI**
- Enterprise knowledge base + policy content repositories
- Backend integrations (customer, order, inventory, etc.)
- Web UI for agents (real-time feed)
- Admin console + dashboards

Copyright © 2026 Torry Harris Business Solutions FZ - LLC

Third-party AI tools used

Microsoft Azure Speech to Text enabled real-time transcription of live conversations, allowing the solution to capture intent and context as interactions unfolded.

Microsoft Azure OpenAI powered the generation of grounded responses, recommendations, and next-best guidance by combining LLM capabilities with retrieved enterprise knowledge and backend data.

What makes it distinctive

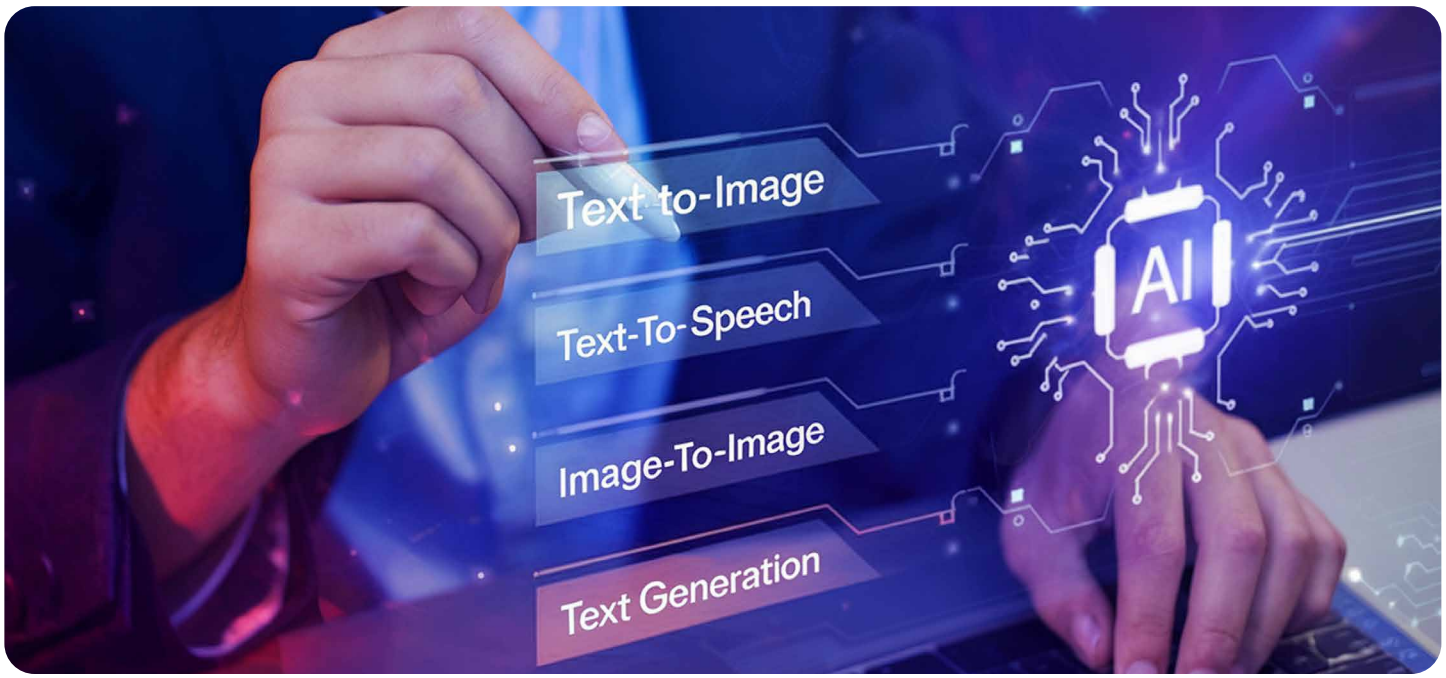
Real-time augmentation:
designed for live calls, not just post-call analysis

Grounded responses:
retrieval from enterprise knowledge + operational context

Integration-friendly:
API-first connectivity to backend sources and channels

Reusable product approach:
modular, extensible components for rapid rollout

Governance-ready:
admin console + dashboards for control and improvement



Copyright © 2026 Torry Harris Business Solutions FZ - LLC

Gen AI augmentation delivers measurable operational gains

By embedding GenAI into the live call workflow, the solution improved resolution speed, reduced transfers, and increased agent productivity by keeping the agent continuously equipped with the most relevant context and guidance.

Business benefits delivered:

Agent efficiency increased by **40%**

AHT reduced by **50%**

Agent utilization increased by **25%**

Resolution rate increased by **45%**

Transfer rate reduced by **70%**

Copyright © 2026 Torry Harris Business Solutions FZ - LLC

About Torry Harris

Established in 1998, Torry Harris Integration Solutions (THIS) is a global digital integration and modernization specialist. Torry Harris helps enterprises build connected, composable architectures using AI, API-led integration, microservices, event-driven patterns, and cloud-native approaches - enabling organizations to modernize legacy estates, streamline operations, and securely build and scale digital ecosystems. Torry Harris is headquartered in New Jersey, USA, with offices in the UK, Ireland, France, Germany, UAE, KSA, and India.

THIS
TORRY HARRIS

For more information, please write to **info@thbs.com**
Website: **www.torryharris.com/ae-en**

All content / information present here is the exclusive property of Torry Harris Business Solutions Private Limited. The content / information contained here is correct at the time of publishing. No material from here may be copied, modified, reproduced, republished, uploaded, transmitted, posted or distributed in any form without prior written permission from Torry Harris. Unauthorized use of the content / information appearing here may violate copyright, trademark and other applicable laws, and could result in criminal or civil penalties.

Copyright © 2026 Torry Harris Business Solutions FZ - LLC